Telecom Customer Churn Analysis & Retention Insights

IBM Data Analytics with AI Internship Project

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Project Type: Data Analytics + Predictive Classification

1. Executive Summary

This project analyzes customer churn for a fictional telecommunications company. The purpose is to transform customer data into actionable business intelligence by identifying churn patterns, customer segments requiring attention, and potential retention opportunities. A Logistic Regression model is also included as a supporting classification exercise.

2. Business Problem

Customer churn can reduce recurring revenue and increase the need to acquire replacement customers. The project asks: Which customer segments show higher churn, which customer/account factors are associated with churn, and how can the business use these insights to prioritize retention activities?

3. Dataset

The project uses the public IBM Telco Customer Churn sample dataset. It contains customer demographics, services, contract and payment information, monthly charges, total charges, tenure, and a Churn outcome. The dataset source is recorded in README.md and in the project code.

4. Data Preparation

• Standardize whitespace in text fields.

• Convert MonthlyCharges and TotalCharges to numeric values.

• Handle blank TotalCharges values.

• Remove duplicate customer IDs.

• Create ServiceCount, TenureBand, and ChurnFlag.

5. Analysis Plan

The analysis follows the internship masterclass framework: KPIs → trends → drivers → risks/opportunities → actions. The main views cover overall churn, contract type, tenure, and payment method. Additional service-level analysis can be extended from the cleaned dataset.

6. Predictive Model

A Logistic Regression classifier is trained with a stratified train/test split. Categorical variables are one-hot encoded, numeric variables are standardized, and class weighting is used to address imbalance. The script saves classification metrics, a confusion matrix, and ROC-AUC in the outputs folder.

7. Business Recommendations Framework

• Monitor the customer segments with the highest observed churn rates.

• Prioritize retention communication for higher-risk segments identified by the analysis.

• Review contract, payment, tenure, and service patterns before designing targeted offers.

• Use the model as a prioritization aid rather than treating predictions as certainty.

8. Outputs / Screenshots

The required charts and analysis outputs are generated automatically by telecom_churn_analysis.py. Run the script once, then insert the four PNG charts from the outputs folder into this report before final submission. This keeps the report tied to the exact run of the submitted code instead of using unverified or manually invented values.

9. Limitations

The dataset is a cross-sectional sample rather than a monthly longitudinal customer history. Therefore, the model should not be described as forecasting a specific future month or proving causation. The project is intended for educational and portfolio use.

10. Submission Files

File	Purpose
telecom_churn_analysis.py	Complete project code
requirements.txt	Python dependencies
Project_Report.pdf	Project documentation/report
README.md	Project overview, dataset source and run instructions
GitHub repository	Repository containing the four files